F70LA-S1 – Life Insurance Mathematics A, 2025/26
Chapter 2 – Survival Models

Read Dickson, Hardy & Waters, Chapter 2.

2.1 Future lifetime random variable

Notation 2.1 (x) *denotes 'a person currently aged* x*'.*

Definition 2.2 *The random variable* T_x *is defined to be the future lifetime of* (x) *from today.*

Note:

- T_x is the time from today until the death of a person who is aged x years today.
- Thus $x + T_x$ is the age when they die.
- T_x is a strictly positive number, but is unknown.
- Thus T_x is a strictly positive real–valued random variable.

The cumulative distribution function of T_x is

$$\boxed{F_x(t) = \mathbb{P}[T_x \leq t] = \mathbb{P}[\ (x) \text{ dies before age } x + t].}$$

The survival function of T_x is

$$\boxed{S_x(t) = 1 - F_x(t) = \mathbb{P}[T_x > t] = \mathbb{P}[\ (x) \text{ survives to age } x + t].}$$

2.2 Consistency condition between T_0 and T_x

Consider a single individual from birth:

- T_0 represents the remaining future lifetime from birth, and hence the age at death.
- If the person survives to age x, then T_x represents the remaining future lifetime from age x, and hence the age at death is $x + T_x$.
- For the survival model to be consistent, the following condition must hold for all $x \geq 0$ and for all $t \geq 0$:

  $$\mathbb{P}[T_x \leq t] = \mathbb{P}[T_0 \leq x + t \mid T_0 > x]$$

  or, equivalently,

  $$\boxed{\mathbb{P}[T_x > t] = \mathbb{P}[T_0 > x + t \mid T_0 > x].}$$

Exercise 2.3 *Show that for* $x, t, u \geq 0$,

$$\boxed{S_x(t) = \frac{S_0(x+t)}{S_0(x)}}.$$

Hence show that

$$S_{x+t}(u) = \frac{S_x(t+u)}{S_x(t)}.$$

2.3 Limiting age

Many survival models have a *limiting*, or maximum, age beyond which it is not possible to live.

- For human mortality, a reasonable limiting age might be 130 years.
- The actuarial notation for the limiting age is ω.
- For a finite limiting age ω, $\mathbb{P}[T_x > \omega - x] = 0$.
- If a finite limiting age does not exist, then $\omega = \infty$ and for all $t \geq 0$, $\mathbb{P}[T_x > t] > 0$. Under a simple exponential mortality model, the limiting age is infinity.

2.4 Conditions to be satisfied by a survival function

The function $S_x(t)$ is a survival function from age $x \geq 0$ if and only if the following three conditions are true:

Number	Condition	Interpretation
(1)	$S_x(0) = 1$	(x) is alive at time 0.
(2)	$\lim_{t \to \omega - x} S_x(t) = 0$, where $\omega \leq \infty$ is the limiting age.	(x) is dead by age ω.
(3)	For $t, u \geq 0$, $S_x(t) \geq S_x(t+u)$, i.e. $S_x(t)$ is a non-increasing function of t.	The probability of survival decreases as (x) gets older.

If $S_x(t)$ is a differentiable function of t, then Condition (3) can be equivalently expressed as $\frac{d}{dt} S_x(t) \leq 0$.

Example 2.4 *An exponential survival model:*

$$S_0(t) = e^{-t}, \quad 0 \leq t < \infty. \qquad \textit{In this model, the implied limiting age } \omega = \infty.$$

A linear survival model:

$$S_0(t) = 1 - \frac{t}{100}, \quad 0 \leq t \leq 100. \qquad \textit{In this model, the implied limiting age } \omega = 100.$$

Exercise 2.5 *Write an expression in terms of the survival function $S_0(t)$ for the probability that a life aged 0 dies between ages 30 and 40.*

Exercise 2.6 *Calculate the probability that a life (0) will die between ages 19 and 36, given the survival function*

$$S_0(x) = \frac{1}{10}\sqrt{100 - x}, \quad 0 \leq x \leq 100.$$

Without being told explicitly, the limiting age is $\omega = 100$.

2.5 Force of Mortality

Definition 2.7 *For a given survival model, the* **force of mortality** *at age x is*

$$\boxed{\mu_x = \lim_{h \to 0} \left\{ \frac{1}{h} \mathbb{P}[0 < T_x \leq h] \right\}.}$$

The force of mortality is effectively the annualised instantaneous rate of death. It is *not* a probability, but can be used to approximate the probability of death.

In the field of engineering, μ_x is called the *hazard rate*. In statistics, it is called the *transition intensity*.

Interpretation of the force of mortality: Let h be a small, positive number. From the definition of μ_x,

$$h \cdot \mu_x \;\approx\; \mathbb{P}[0 < T_x \leq h] = \mathbb{P}[(x) \text{ dies before age } x + h\ ].$$

For example, suppose $\mu_{80} = 0.1$. Then the approximate probability of a 80-year-old dying during the next month (i.e. choose $h = 1/12$ years) is $\frac{1}{12}\mu_{80} = 0.01$. Note that μ_x is generally continually increasing at older ages x. This means that $\mu_{80\frac{1}{12}} > \mu_{80}$.

Example 2.8 *Show that:*

(a)

$$\boxed{\mu_x = -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x) = -\frac{d}{dx} \ln S_0(x).}$$

(b)

(1) $$\boxed{S_x(t) = \exp\left(-\int_x^{x+t} \mu_y \, dy \right) = \exp\left(-\int_0^t \mu_{x+u} \, du \right).}$$

(c)

$$\boxed{\mu_{x+t} = -\frac{1}{S_x(t)} \frac{d}{dt} S_x(t) = -\frac{d}{dt} \ln S_x(t).}$$

(a)

$$\begin{aligned}\mu_x = \lim_{h\to 0}\left\{\frac{1}{h}\mathbb{P}[0 < T_x \le h]\right\} &= \lim_{h\to 0}\left\{\frac{1}{h}\frac{S_0(x) - S_0(x+h)}{S_0(x)}\right\} \\ &= -\frac{1}{S_0(x)}\frac{d}{dx}S_0(x) \\ &= -\frac{d}{dx}\ln S_0(x).\end{aligned}$$

(b) From part (a),

$$\begin{aligned}\int_x^{x+t}\mu_y dy = \int_x^{x+t}\left(-\frac{d}{dy}\ln S_0(y)\right)dy &= \ln S_0(x) - \ln S_0(x+t) \\ &= \ln\left(\frac{S_0(x)}{S_0(x+t)}\right) \\ &= -\ln\left(\frac{S_0(x+t)}{S_0(x)}\right) \\ &= -\ln S_x(t),\end{aligned}$$

where the last line follows from the relationship in Example 2.3. Thus

$$S_x(t) = \exp\left(-\int_x^{x+t}\mu_y dy\right) = \exp\left(-\int_0^t \mu_{x+s}ds\right).$$

(c) Now fix $x \ge 0$ and let t vary. From part (b),

$$\frac{d}{dt}S_x(t) = -\mu_{x+t}\,\exp\left(-\int_0^t \mu_{x+s}ds\right) = -\mu_{x+t}\,S_x(t).$$

Hence

$$\mu_{x+t} = -\frac{1}{S_x(t)}\frac{d}{dt}S_x(t) = -\frac{d}{dt}\ln S_x(t).$$

Important points

- If we know $S_0(x)$ for all $x \geq 0$, then we can calculate $S_x(t)$ for all $x, t \geq 0$. How? By

$$S_x(t) = \frac{S_0(x+t)}{S_0(x)},$$

which shows that the survival probabilities from age 0 determine all the future survival probabilities.

- If we know the force of mortality μ_x for all $x \geq 0$, then we can calculate the survival function $S_x(t)$ for all $x, t \geq 0$. How? By using the relationship

$$S_x(t) = \exp\left(-\int_0^t \mu_{x+s} ds\right),$$

which shows that $S_x(t)$ is a function of all the forces of mortality from age x up to age $x+t$.

2.6 International actuarial notation

${}_tp_x := S_x(t) = \mathbb{P}[\ (x) \text{ survives to age } x+t]$	Said: "t p x"
${}_tq_x := F_x(t) = \mathbb{P}[\ (x) \text{ dies before age } x+t]$	Said: "t q x"

Exercise 2.9 *Show that*

$$\boxed{{}_tp_x + {}_tq_x = 1.}$$

$${}_{t|u}q_x = \mathbb{P}[(x) \text{ dies between ages } x+t \text{ and } x+t+u] \qquad \text{Said: "}t \text{ deferred } u \text{ } q \text{ } x\text{"}$$
$$= \frac{S_0(x+t) - S_0(x+t+u)}{S_0(x)} = \frac{S_0(x+t)}{S_0(x)}\left(1 - \frac{S_0(x+t+u)}{S_0(x+t)}\right),$$

i.e.

$$\boxed{{}_{t|u}q_x = {}_tp_x - {}_{t+u}p_x = {}_tp_x \cdot {}_uq_{x+t}.}$$

For consistency between the deferred mortality probability ${}_{t|u}q_x$ and the mortality probability ${}_tq_x$, it holds that when the deferral period is 0 years, ${}_{0|t}q_x = {}_tq_x$.

When dealing with one-year survival and mortality probabilities, the following notation is generally used:

$$\boxed{p_x \equiv {}_1p_x, \qquad q_x \equiv {}_1q_x, \qquad {}_{t|1}q_x = {}_{t|}q_x.}$$

Exercise 2.10 *Show that survival probabilities are multiplicative, i.e. for* $0 \leq s \leq t$,

$$\boxed{{}_tp_x = {}_sp_x \, {}_{t-s}p_{x+s}.}$$

Hence show that for $k = 1, 2, \ldots$,

$${}_kp_x = p_x \, p_{x+1} \, p_{x+2} \cdots p_{x+k-1} = \prod_{m=0}^{k-1} p_{x+m}.$$

Exercise 2.11 *Show that:*

(a) the PDF of T_x satisfies

$$\boxed{f_x(t) := \frac{d}{dt}F_x(t) = {}_tp_x \cdot \mu_{x+t}}.$$

Hint: Use that $F_x(t) = 1 - S_x(t)$ and differentiate (1) shown in Example 2.8.

(b) the CDF of T_x satisfies,

$$(2) \qquad \boxed{F_x(t) = {}_tq_x = \int_0^t {}_sp_x \cdot \mu_{x+s}\, ds}.$$

(c) ${}_{t|u}q_x = \int_0^u {}_{s+t}p_x \cdot \mu_{x+t+s}\, ds.$

2.7 Some Laws of Mortality

- Gompertz' Law (1825): Given constants $B > 0$ and $c > 1$, the force of mortality at age $x \geq 0$ is

$$\mu_x = Bc^x.$$

- Makeham's Law (1860): Given constants $A, B > 0$ and $c > 1$, the force of mortality at age $x \geq 0$ is

$$\mu_x = A + Bc^x.$$

Exercise 2.12 *(a) Show that under Gompertz' Law,*

$${}_tp_x = g^{c^x(c^t-1)},$$

where the constant $g = \exp(-B/\ln c)$.

(b) Show that under Makeham's Law,

$${}_tp_x = s^t\, g^{c^x(c^t-1)},$$

where the constant $s = e^{-A}$ *and* $g = \exp(-B/\ln c)$.

2.8 Mean and standard deviation of T_x

We want expressions for $\mathbb{E}(T_x)$ and $\sqrt{\mathbb{V}ar(T_x)}$. To be able to calculate these, the following technical assumptions are sufficient. They are satisfied by all reasonable survival functions.

$$\lim_{t\to(\omega-x)} \{t\,{}_tp_x\} = 0 \qquad \text{and} \qquad \lim_{t\to(\omega-x)} \left\{t^2\,{}_tp_x\right\} = 0.$$

Notation 2.13 *The expected future lifetime from age x is* $\boxed{\mathring{e}_x := \mathbb{E}(T_x)}$.

Exercise 2.14 *Justify each step in the following calculation:*

$$\begin{aligned}\mathring{e}_x &= \int_0^{\omega-x} t\frac{d}{dt}F_x(t)\,dt = \int_0^{\omega-x} t\frac{d}{dt}(1-S_x(t))\,dt = -\int_0^{\omega-x} t\frac{d}{dt}S_x(t)\,dt \\ &= -\int_0^{\omega-x} t\,\frac{d}{dt}({}_tp_x)\,dt \\ &= -\left[t\cdot {}_tp_x\right]_0^{\omega-x} + \int_0^{\omega-x} {}_tp_x\,dt.\end{aligned}$$

Hence, by the first technical assumption,

$$\boxed{\mathring{e}_x = \int_0^{\omega-x} {}_tp_x\,dt.}$$

Exercise 2.15 *Show that*

$$\mathbb{V}ar(T_x) = 2\int_0^{\omega-x} t\,{}_tp_x\,dt - \left(\int_0^{\omega-x} {}_tp_x\,dt\right)^2.$$

Exercise 2.16 *Suppose the CDF of* T_0 *is*

$$F_0(t) = 1 - (1 - t/120)^{1/6}, \quad \text{for } 0 \leq t \leq 120.$$

(a) For age $x \in [0, 120)$ *and* $t \in [0, 120 - x)$*, derive an expression for* $S_x t$ *in terms of* x *and* t *only.*

(b) Calculate $\mathring{e}_{30}$*.*

(c) Calculate the standard deviation $\mathbb{SD}(T_{30})$*.*

2.9 Curtate future lifetime random variable K_x

Definition 2.17 *The number of complete years of future lifetime of (x) is called the* **curtate future lifetime of (x)**. *It is denoted by the random variable K_x, with*

$K_x :=$ *the integer part of* T_x.

Note that K_x is a discrete random variable, which takes integer values $0, 1, 2, \ldots$.

Example 2.18 *If in some future state of the world ω, a life* (50) *dies at age* 95.3 *years. Then $K_{50}(\omega) = 45$, while $T_{50}(\omega) = 45.3$.*

The probability mass function of K_x is

$$\boxed{\mathbb{P}[K_x = k] = \mathbb{P}[k \leq T_x < k+1] = \mathbb{P}[(x) \text{ dies between ages } x+k \text{ and } x+k+1] = {}_{k|}q_x.}$$

Note:

- The life (x) must die at some future time, so

$$\sum_{k=0}^{\infty} \mathbb{P}[K_x = k] = \sum_{k=0}^{\infty} {}_{k|}q_x = 1.$$

- If the maximum future lifetime $\omega - x$ is a positive integer, then

$$\sum_{k=0}^{\omega-x-1} \mathbb{P}[K_x = k] = \sum_{k=0}^{\omega-x-1} {}_{k|}q_x = 1.$$

Notation 2.19 *The curtate expectation of life from age x is* $\boxed{e_x := \mathbb{E}(K_x)}$.

Exercise 2.20 *Show that*

$$e_x = \sum_{k=1}^{\infty} {}_kp_x.$$

Exercise 2.21 *(a) Explain why* $\mathbb{E}[T_x \mid T_x > 1] = 1 + \mathring{e}_{x+1}$.

(b) Show that $\mathring{e}_x \leq \mathring{e}_{x+1} + 1$.

(c) Show that $\mathring{e}_x \geq e_x$.

(d) Explain why $\mathring{e}_x \approx e_x + \frac{1}{2}$ *might be a reasonable approximation.*

(e) Is $\mathring{e}_x$ *always a non-increasing function of* x*?*